



COURSE GUIDE BOOK FOR HEAT TRANSFER

1. GENERAL INFORMATION OF THE COURSE

Course Name:	Heat transfer
Course Code:	MEng 3172
Course Credit Points	5hrs.
Contact Hours:	2 hrs.lecture,2 hrs. Tutorial & 1hrs.laboratoryor practice per week
Course Module:	Eng'g Thermo-fluids
Course Category:	Major
Status of Course:	Compulsory
Pre-Requisites:	Applied Mathematics III, Engineering Thermodynamics I and II
Students' Department:	Mechanical Engineering
Degree Program:	BSC Mechanical Engineering
Students' Admission:	Regular
Year of Study:	3 rd .
Semester:	Second Semester
Academic Year:	2024/2025
Course Duration:	From March2024toJune 2024
Course Chair:	Abraham Adera
Course Instructors:	Senait Tadesse,MSc Mechanical Engineering Email:senaitnese @gmail.com Office:NDD 203
Consultation:	Monday 9:30AM-11:00AM Friday 9:30AM-11:00AM

2. COURSE DESCRIPTION

The course is designed to familiarize the students with the basic principles of heat and mass transfer mechanisms and applications. Students will learn in detail

- (i) The basic concepts of heat transfer, conduction, convection and radiation,
- (ii) How their combinations contribute in any heat transfer process, and how a heat transfer process can be made more efficient and how to reduce heat losses.

It would also address various types of heat exchangers, their analysis, selection and sizing; basics of natural and forced convection, black body radiation, estimation of radiative heat exchange between two

hot black and gray bodies. The mass transfer mechanism which, from analysis point of view is very similar to heat transfer mechanism but very different applications has also been included in the paper discussed. Basic concepts related to mass transfer through diffusion and mass convection have been dealt in details. Application of mass transfer simultaneously along with heat transfer in various unit processes of evaporation, distillation, absorption, adsorption, humidification, dehumidification, drying etc. has also been described.

3. OBJECTIVES OF THE COURSE

The specific objectives the course are:

- Solve steady and Transient, one-dimensional conduction heat transfer problems analytically
- Use Numerical/finite difference approaches to numerically solve two dimensional, conduction and transient heat transfer problems
- Understand fins and analyze heat transfer from extended surfaces
- Use methods available for analysis of transient heat transfer problems
- Understand the important of dimensionless parameters to convection heat transfer and basic principles underlying convection heat transfer
- Understand thermal and viscous boundary layer heat transfer phenomena for laminar and turbulent flows
- Analyzing heat transfer in various flow conditions (internal and external flows) by determining convection coefficient for a flow in/on a particular geometry.

4. Student Learning Outcome

Students who successfully complete this course will be able to:

1. Develop the fundamental heat transfer equations in Cartesian, cylindrical and spherical coordinates.
2. Develop the students' abilities to model and analyze thermal systems.
3. Develop experience in the application of thermal analysis to elementary problems in engineering practice

1. The Course Time Schedule and Teaching Methodology

The following topics will be covered in this course within this semester.

Table 1. Time Schedule and Teaching Methodology of the course

Time Schedule	Course Contents	Teaching and learning Methodology	Attaining student outcome
Week 1	UNIT 1 1. Introduction to Heat & Mass Transfer 1.1. Conduction Heat Transfer 1.2. Convective Heat Transfer 1.3. Radiation Heat Transfer 1.4. Governing Equations of Heat Conduction 1.5. Diffusion Mass Transfer	• Lecture, tutorial discussions • Two example • Text book1,Ref2	Outcome: 1
Week 2-5	UNIT 2 2. One Dimensional Steady State Conduction 2.1. The Heat Diffusion Equation 2.2. The Plane Wall 2.3. Thermal Resistance and The Overall Heat Transfer Coefficient 2.4. Radial Systems 2.5. Conduction with Thermal Energy Generation 2.6. Heat Transfer from Extended Surfaces	• Lecture, tutorial discussions • lab • four example • Text book1,Ref2	Outcome: 1,2

Week 6-7	UNIT 3 3. Two-Dimensional Steady State Conduction 3.1. Mathematical Analysis 3.2. Finite Difference Method	- Lecture, tutorial discussions - Four example Text book1,Ref2	Outcome: 1,2,3,
Week 8	Mid exam		
Week 9	UNIT 4 4. Transient Heat Conduction: 4.1. The lumped capacitance method; Validity of lumped capacitance method; 4.2. One-dimensional system with convective surface conditions (application of Heisler and Grober Charts); 4.3. Graphical method - Schmidt Plot	- Lecture, tutorial discussions - three example - Text book1,Ref2	Outcome: 2,3
Week 10	UNIT 5 5. Introduction to convection: 5.1. The convection transfer equation; 5.2. Convection boundary layers: Velocity boundary layer and thermal boundary layer; 5.3. Significance of boundary layer; Laminar and turbulent flow	- Lecture, tutorial discussions - Text book1,Ref2	Outcome: 1,2,3
Week 11	UNIT 6 6. Forced Convection: 6.1. Methodology for convection calculation of flat plate in parallel flow with laminar; 6.2. Mixed and turbulent flow conditions; The cylinder in cross flow; The sphere; Flow across bank of tubes; Internal flow; 6.3. The mean velocity; Velocity profile in fully developed region; 6.4. The mean temperature; Convection correlation laminar flow in circular tubes for fully developed	- Lecture, tutorial discussions - three example - Text book1,Ref2	Outcome: 1,2,3,4

	and entry regions; 6.5. Convective correlation for turbulent flow in circular tubes; Convection correlation for non-circular tubes.		
Week 12	UNIT 7 7. Free Convection: 7.1. Physical considerations; 7.2. The governing equation; Free convection on a vertical plate; 7.3. Empirical correlations for external flows of vertical- plate, inclined and horizontal plates.	• Lecture, tutorial discussions • Three example • Text book1,Ref2	Outcome: 1,2,3,
Week 13	UNIT 8 8. Heat Transfer with phase change: 8.1. Condensation and Boiling Heat Transfer 8.2. Boiling modes 8.3. Condensation mechanisms	• Lecture, tutorial discussions • three example • Text book1,Ref2	Outcome: 1,2,
Week 14	Unit 9 9. Heat Exchangers: 9.1 Heat exchanger types; The overall heat transfer coefficient; 9.2 Heat exchanger analysis using log mean temperature difference and the effectiveness-NTU method; 9.1. Method of heat exchanger calculation	• Lecture, tutorial discussions • four example • lab • Text book1,Ref2	Outcome: 2,3,4
Week 15	Unit 10 10. Radiation Heat Transfer: 10.1. Blackbody radiation exchange; 10.2. Radiation exchange at surface; The view factor; View factor relations; 10.3. Radiation exchange between surfaces.	• Lecture, tutorial discussions • Four example • Text book1,Ref2	Outcome: 1,2,,4
Week 16	Final Exam		

5. Teaching & Learning Methods

Lecture supported by laboratory exercises

6. Assessment and Evaluation

The course utilizes a variety of continuous assessment and evaluation methods to ensure the successful achievement of course learning outcomes. These methods include assignments, projects, quizzes, midterm exams, final exams, site visit reports, homework, classwork, presentations, and more. It is imperative that all assignments and tasks are completed within the specified deadlines. If you encounter any difficulties with the assignments or tests, please contact the instructor promptly for assistance.

Home works, Assignments and project: Assignments may involve collaborative work for brainstorming and developing general approaches. However, unless specified otherwise, each individual must complete their own submissions. Assignments and projects may entail both individual and group components, with equal participation expected from all group members. While collaboration is encouraged, submitting another individual's work as one's own constitutes academic dishonesty and will incur in penalties. All group members must contribute equally. Timely submission is mandatory, as late submissions will not be graded. Oral exams will accompany assignments and projects, requiring thorough preparation from all members. Additionally, the instructor reserves the right to assign additional tasks or assignments as necessary.

Quizzes: Quizzes serve to assess students' current understanding of the coursework. They are conducted in class following lecture or

tutorial sessions.

Mid-term Exam and Final Exam: The exams will be closed book. During exam all necessary materials will be provided by examiner.

Table 2. Summary of Assessments

No	Assessment type	Mark allotted	Content Coverage	Attaining course learning outcome (CLO)	Time Schedule
1	Individual Assignment 1	5%	Chapter 1,2&3	Outcome: 1,2	Week 3
2	Quiz Exam 1	10%	Chapter 1,2 & 3	Outcome: 1,2	Week 4
3	lab report	10%			Week 6,14
4	Mid exam	25%	Chapter 1,2,3 &4	Outcome: 1,2,3,4	Week 8
5	Final Exam	50%	Chapter 5,6,7,8,9&10	Outcome: 1,2,3,4	Week 16
	Total	100%			

7. Laboratory/Workshop/ session

Table 3. Summary of laboratory /Work shop/ session content and required material

No	Schedule	Laboratory Topic	Short Description of laboratory	Attaining course outcome
1	Week 1	Heat conduction	Linear and radial heat conduction	
2	Week 2	Heat exchanger	Shell tube heat exchanger performance under parallel flow and counter flow	
3	Week 3	Heat exchanger	Investigation of pressure drop characteristic of a finned tube heat exchanger	

8. COURSE POLICY

All students are expected to adhere to the code of conduct for students as per the Senate Legislation of Bahir Dar University, December 2020, during the duration of this course. Academic misconduct, which includes cheating, fabrication, and plagiarism, will not be tolerated and will be promptly reported to the relevant authorities for appropriate action.

The teaching methodology for this course will be diverse, encompassing activities such as reading, lectures, tutorials, construction site visit, discussions, homework, classwork exercises, and presentations, etc. Active participation from students is expected, including asking questions and raising issues whenever clarification is needed.

Regular attendance and punctuality are expected from all students. Those who miss more than 15% of the semester's classes will not be eligible to sit for the final exam. Attendance will be recorded for each class, and students are responsible for all announcements and changes made in class. Any student who exceeds the 15% absence threshold will receive a grade of 'NG' (No Grade) and must provide acceptable reasons for their absence, as per Senate legislation, Art. 174 for NG management.

Cell phones brought to class must be set to silent mode to avoid disruption to the teaching-learning process. Any form of disturbance during class time is unacceptable and must be avoided.

All students are required to have complete records of course assessments and evaluations. If a student misses a quiz, midterm exam, or any other continuous assessment without a valid reason, they will receive a score of zero. However, if a student fails to attend the final exam or complete a term paper/project without a valid reason, the instructor will record an "NG" according to senate legislation, Art. 171.

It is important to carefully follow the instructions provided in your course guidebook for completing all requirement of the course.

9. Grading system

The grading scale of the course is fixed scale according to senate legislation of Art. 177.

Table 4. Grading scale of the course

Raw Mark Interval (100%)	Corresponding Letter Grade	Corresponding Fixed Number Grade
$90 \leq x \leq 100$	A ⁺	4.0
$85 \leq x < 90$	A	4.0
$80 \leq x < 85$	A ⁻	3.75
$75 \leq x < 80$	B ⁺	3.50
$70 \leq x < 75$	B	3.00
$65 \leq x < 70$	B ⁻	2.75
$60 \leq x < 65$	C ⁺	2.50

$50 \leq x < 60$	C	2.00
$45 \leq x < 50$	C-	1.75
$40 \leq x < 45$	D	1.00
$x < 40$	F	0.00

10. LITERATURE

TEXTBOOK:

1. Fundamentals of Heat and Mass Transfer Incropera/ DeWitt/ Bergman/ Lavine, 6th Edition

References:

1. Frank P. Incropera and David P. DeWitt, Fundamentals of Heat and Mass Transfer, 5th Edition, Aug 9, 2001.
2. Yunus A. Cengel, Heat and Mass Transfer: A Practical Approach w/ EES CD, Jan 4, 2006.
3. Holman J P, Heat Transfer, Oct 10, 2001.
4. Heat Transfer - A. J. Chapman
5. Heat Transfer - Eckert and Drake
6. Engineering Heat Transfer - C. P. Gupta

The following bodies confirmed the course guidebook accordingly.

Position	Name	Signature	Date
Course Instructors	Senait Taddese		
Course Module Chair	Abraham Adera		
Chair Holder (Thermal Engineering)	Dr. Bimrew Tamrat		
v/dean	Abebe Genetu		